

Determining Inventor Status and Its Effect on Knowledge Diffusion: A Study on Nanotechnology Literature From China, Russia, and India

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In an increasingly global research landscape, it is important to identify the most prolific researchers in various institutions and their influence on the diffusion of knowledge. Knowledge diffusion within institutions is influenced by not just the status of individual researchers but also the collaborative culture that determines status. There are various methods to measure individual status, but few studies have compared them or explored the possible effects of different cultures on the status measures. In this article, we examine knowledge diffusion within science and technology-oriented research organizations. Using social network analysis metrics to measure individual status in large-scale coauthorship networks, we studied an individual's impact on the recombination of knowledge to produce innovation in nanotechnology. Data from the most productive and high-impact institutions in China (Chinese Academy of Sciences), Russia (Russian Academy of Sciences), and India (Indian Institutes of Technology) were used. We found that boundary-spanning individuals influenced knowledge diffusion in all countries. However, our results also indicate that cultural and institutional differences may influence knowledge diffusion.

Introduction

Scientific publications are important indicators of innovative research, which has been defined as discovering solutions to problems via the novel recombination of existing knowledge (Fleming, 2001). The diffusion of existing knowledge can be traced by this recombinatory process and the factors that influence it. Researchers have a choice in selecting knowledge that is recombined to produce new innovations, and studies have suggested that this choice is influenced by the status of individuals within knowledge networks (Nerkar & Paruchuri, 2005; Singh, 2007). Individuals usually value and recombine innovations from high-status researchers.

Previous research has used various social network analysis (SNA) measures to establish the status of individuals and examine their role on knowledge diffusion. Patterns of collaboration at various departmental, organizational, and national granularities also have been studied. However, there has been limited comparative examination of social network measures to establish individual status. In addition, there has been little research on whether cultural differences influence individual status and the recombinatory knowledge-diffusion process. We believe that in an increasingly competitive global research landscape, identifying the most prolific individuals in various countries and disciplines based on the most effective measures is important.

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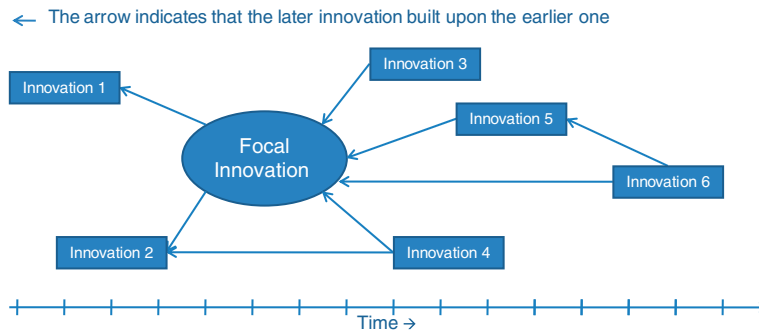


FIG. 1. Emergence of a focal innovation.

In this study, we extract large-scale coauthorship networks from the most prolific institutions in China, Russia, and India in the fast-moving nanotechnology discipline. We examine an individual researcher's status and its impact on knowledge diffusion. We also explore if the collaborative cultures in these countries have an effect on the effectiveness of status measures. Specifically, the research questions explored are:

- RQ1:** How does the status of an individual in a coauthorship network affect knowledge diffusion?
- RQ2:** Which measures can explain the effect of an individual's status on knowledge diffusion?
- RQ3:** Is the measure of author status in knowledge networks different across countries and institutional cultures?

The next section presents the literature review. The subsequent section presents the research design and testbed used in our work. The experiment results are then shown and discussed. Last, we conclude and present possible future directions in this area.

Literature Review

In this section, we discuss important previous research on coauthorship-based knowledge networks. We characterize knowledge networks as social networks with authors as nodes and coauthorship collaborations as links between them. A summary of the social network measures to determine individual status also is presented.

Knowledge Networks

Since the first collaborative scientific paper was published, coauthorship has been increasing steadily over time across all fields. Innovation by an individual in a knowledge network is greatly influenced by his or her collaboration and interaction patterns. To understand the diffusion of knowledge in these networks, it is necessary to identify the factors leading to the selection of a "focal innovation." Podolny and Stuart (1995) defined the focal innovation as the knowledge that others build upon to produce novel outcomes. Figure 1 shows the process of evolution of knowledge (time increasing on the x -axis) and the emergence of a focal innovation. An arrow from

one innovation to the next indicates that the second innovation built upon the first. This process of knowledge diffusion can be traced using citations in research literature. Indeed, citation analysis has been widely used in coauthorship networks to study various network and diffusion phenomena (Bornmann & Daniel, 2008; Davis, 2009; Leydesdorff & Sun, 2009).

Knowledge networks can be studied at different levels of granularity, with emphasis on individual, within-group, or intergroup collaboration. Previous studies have examined the patterns of collaboration between individual authors in various disciplines (Kretschmer, 2004; Moody, 2004; Newman, 2001; Newman, 2004; Yoshikane & Kageura, 2004). Other research also has focused on within-group collaboration (Arunachalam & Doss, 2000; Arunachalam, Srinivasan, & Raman, 1994; Luukkonen, Persson, & Sivertsen, 1992; Zitt, Bassecouard, & Okubo, 2000) and intergroup collaboration (Leydesdorff & Sun, 2009; Singh, 2007; Wagner & Leydesdorff, 2005; Yamashita & Okubo, 2006; Zucker & Darby, 1996).

Though there is a difference in granularity, the primary research focus of previous studies are twofold: (a) network structure, which analyzes the topological structure, its evolution, and the effect of global measures on knowledge generation, diffusion, and recombination within or across the organizations; and (b) individual status, which analyzes trends or patterns in an individual's collaborative activity and links to determine status. Table 1 presents a representative set of these studies.

As can be seen from Table 1, many studies have used measures to examine the topological structure of coauthorship networks. Among the limited papers focusing on effect of individual status on knowledge selection, Zucker and Darby (1996) focused on collaboration patterns for star scientists across groups, and results showed that these scientists tend to collaborate more within their own institutions in the early years of their evolution to protect their techniques, ideas, and discoveries. Newman (2001) used the betweenness centrality measure to study individual activities and demonstrated that the typical distances between pairs of authors through the networks are small. Kretschmer (2004) studied the relationship between the average geodesic distances for collaboration clusters and the productivity of their scientists, and found that highly productive authors have, on average, shorter length

TABLE 1. Previous knowledge network studies.

	Study granularity	Study emphasis	Measures	Aim/result
Newman, 2001	Individual	Network structure/ Individual status	Closeness centrality, betweenness centrality	(a) Typical distances between pairs of authors in the networks are small (small world). (b) The bulk of the paths between authors go through just one or two of their collaborators.
Kretschmer, 2004	Individual	Individual status	Closeness centrality: (based on geodesic distance)	Exploration of relationship between (a) size of clusters and (b) average of geodesic distances on the productivity of scientists
Moody, 2004	Individual	Network structure	Degree distribution, clustering coefficient	Impact of social-interaction pattern on the structure of ideas
Newman, 2004	Individual	Network structure	Largest component, largest distance, clustering coefficient	Patterns of collaboration in different disciplines
Yoshikane & Kageura, 2004	Individual	Network structure	No. of partners, Gini's index	Development of personal collaboration networks in different disciplines
Ding, Yan, Frazho, & Caverlee, 2009	Individual	Individual status	PageRank, Weighted PageRank	Compare PageRank and weighted PageRank results with the citation ranking, h-index, and centrality measures. Results: Citation rank is highly correlated with PageRank and weighted PageRank.
Luukkonen, Persson, & Sivertsen, 1992	Intragroup	Network structure	Observed/expected ratios, percentage of papers with international institutional coauthorships	Institutional coauthorship patterns in different countries
Arunachalam, Srinivasan, & Raman, 1994	Intragroup	Network structure	Affinity index, cooperation index (Salton's measure)	Study the international cooperative behavior in Asia giants (e.g., China, Japan, and India).
Arunachalam, & Doss, 2000	Intragroup	Network structure	International cooperation index and affinity index	International collaboration activity in science in 11 different countries in Asia
Zitt, Bassecouard, & Okubo, 2000	Intragroup	Network structure	Probabilistic affinity index	Cooperation behavior in five large scientific countries
Zucker & Darby, 1996	Intergroup	Individual status	Degree centrality	Collaboration patterns for star scientists across groups
Wagner & Leydesdorff, 2005	Intergroup	Network structure	Clustering, links between pairs of countries using Salton's measure and Jaccard's measure	Global network has expanded and become more interconnected. Large countries compete with each other for partners.
Yamashita & Okubo, 2006	Intergroup	Network structure	Probabilistic partnership Index	International collaborative patterns and domestic collaborative patterns within French-Japanese cooperation
Singh, 2007	Intergroup	Individual status	Degree centrality and extensions	Impact of external collaboration on innovation and productivity Result: External collaboration cross-national, organizational, or institutional boundaries increases the productivity

Note. Studies marked in bold specifically focused on the effect of individual status on innovation selection. These are discussed in detail in the text.

of paths to all the other authors. Singh (2007) focused on the impact of external collaboration on innovation and productivity, and found that the degree centrality of an inventor did not have a significant effect on the impact of his or her innovation while the external-collaboration cross national, organizational, or institutional boundaries increased productivity. Ding, Yan, Frazho, and Caverlee (2009) proposed a weighed PageRank algorithm to study how varied damping factors in this algorithm influence the ranking of authors in co-citation network; using an empirical study, they concluded that citation rank is highly correlated with PageRank

with different damping factors and with a different weighted PageRank algorithm. The article also compared PageRank and weighted PageRank results with other measures such as ranking, h-index, and centrality measures; however, it investigated individual status from a static point of view, without tracing the knowledge diffusion and selection process in a dynamic citation network.

From the literature, we found that most previous studies have focused on the macro level, emphasizing intragroup or intergroup collaboration rather than the individual status. There are two primary drawbacks in studies that have

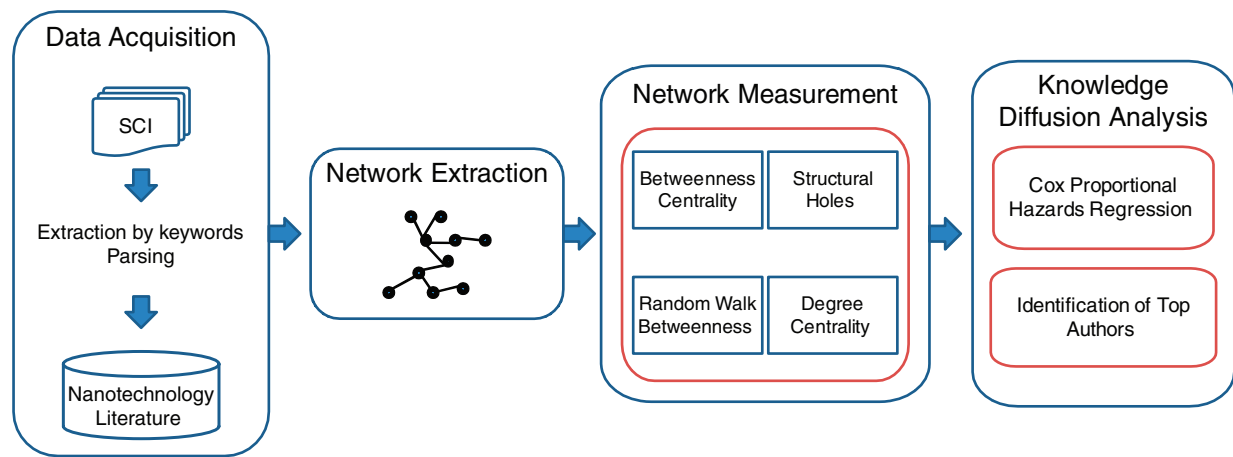


FIG. 2. Research design.

examined individual status. First, most have used static snapshots of networks without taking time into consideration. Since the status of an individual changes over time, it plays an important role in determining the influence on knowledge diffusion. Second, few studies have compared the measures with each other to determine the most effective ones. In addition, there is a need to study the effectiveness of individual measures in different organizations and cultures. In this article, we use a time-dependent analytic technique to measure the status of individuals in a network and study its influence on the citation of their knowledge in datasets from three countries.

Social Network Measures

Various measures to identify high-status nodes have been proposed in prior work (Bonacich, 1987; Burt, 1992; Freeman, 1979; Newman, 2005). These measures are usually known as *centrality* or *prestige* measures and can be broadly classified into three categories based on their methods of calculation: degree centrality, closeness centrality, and betweenness centrality. The degree centrality measures refer to those emphasizing the immediate ties of a node and its neighbors and include the Freeman's degree centrality (Freeman, 1979), Bonacich power (Bonacich, 1987), and structural holes (Burt, 1992), among others. Closeness centrality (Freeman, 1979) emphasizes the distance of a node to all others in the network. Betweenness centrality views a node as being in a favored position to the extent that it falls on paths between other pairs of nodes in the network. Primary betweenness centrality measures include Freeman's betweenness centrality (Freeman, 1979) and the more recent random-walk betweenness centrality (Newman, 2005). Each of these measures makes different assumptions about the flow of knowledge in a network. For instance, degree-based measures emphasize direct links, closeness-based measures are based on the speed of transfer, and betweenness measures emphasize access to diverse parts of the network. Thus, picking the right measure to determine individual status becomes paramount, and comparison becomes important.

Previous studies have analyzed the effect of individual status and knowledge diffusion based on two theories:

Centrality and its impact on knowledge diffusion: Centrality is usually regarded as a signal of a high level of collaboration within the network. Previous empirical studies have indicated that an individual's centrality measured by degree centrality has a positive effect on the diffusion of his or her knowledge (Patrakosol & Olson, 2007; Podolny & Stuart, 1995). In addition, separate studies have shown that Bonacich power and betweenness centrality also had a positive effect (Nerkar & Paruchuri, 2005; Newman, 2001).

Boundary spanning and access to diverse knowledge: Individuals who perform boundary-spanning roles are considered influential as gatekeepers to diverse knowledge bases. Previous empirical studies have provided mixed results: Nerkar and Paruchuri (2005) used the structural holes measure to determine centrality and found that a higher centrality has a positive effect on knowledge diffusion; in another study on structural holes, it was found to have negative effect on innovation (Ahuja, 2000).

However, none of the previous studies have compared these measures to determine which is more effective. In addition, coauthorship networks have not been studied to see the effect (if any) of organizational or cultural environments on the determination of individual status. To address these concerns, we implemented degree centrality (Freeman, 1979), structural holes (Burt, 1992), and betweenness centrality (Freeman, 1979) to model knowledge diffusion patterns in coauthorship networks based on their use in previous literature. In addition, random walk betweenness (Newman, 2005) also is included since this measure is largely different in basis because it assumes random knowledge diffusion and has rarely been used to study knowledge networks.

Research Design

Figure 2 shows the system design and research process used for data acquisition, network extraction, network measurement, and knowledge diffusion analysis.

TABLE 2. Basic statistics of the three institutions under study.

Institutions	Papers	Authors (distinct)	Publication year
CAS	5,881	6,738	1990–2004
RAS	4,055	8,628	1991–2004
IIT	740	1,074	1992–2004

CAS = the Chinese Academy of Sciences; RAS = the Russian Academy of Sciences; IIT = the Indian Institutes of Technology.

Data Acquisition and Research Testbed

We selected Thomson Reuters' *Science Citation Index* (SCI) as our database because it provides broad coverage of journals in more than 150 disciplines, which can be used to extract coauthorship networks and citation information. We narrowed our focus to nanotechnology-related literature, as it promises to impact various engineering and computing disciplines and forms an ideal testbed for a large-scale study on global research outputs. The keyword-searching method was used to collect nanotechnology papers (for more details, see Huang, Chen, Chen, & Roco, 2004). Since SCI papers are in XML format, we developed parsers to extract relevant authorship and citation data. Our final testbed spans from 1976 to 2004, with a total of 213,847 papers published by 231,561 individuals.

We focused on China, Russia, and India because these countries are active in paper publications and are having an increasing impact on global research, especially in nanotechnology (Huang et al., 2004; Li et al., 2008; Liu et al., 2009). China's nanotechnology R&D investment is estimated to have been about 250 million U.S. dollars in 2008 (Guan & Ma, 2007). In Russia, it has been reported that approximately 20 to 25 million U.S. dollars in funding have been invested in nanotechnology research annually since 2002 (Tretyakov, 2007). In India, the government planned to spend 23 million U.S. dollars on nanotechnology between 2004 and 2009, and the investments' new research programs and centers increased significantly after 2007 (Hassan, 2005).

SCI papers also serve as a good testbed to study the effect of different collaborative cultures and work environments on knowledge diffusion. Since we focus on the status of individuals within organizations, we selected the Chinese Academy of Sciences (CAS), the Russian Academy of Sciences (RAS), and the Indian Institutes of Technology (IIT) as our test group. These were picked because they are the most prestigious institutions within the target countries and are the most productive in nanotechnology research (Liu et al., 2009). Table 2 shows the basic statistics of publications in nanotechnology for these organizations.

Network Extraction

A knowledge network based on coauthorships was extracted for authors from all three institutions. The networks were constructed such that each node is an author, and two authors are linked to each other if they have coauthored a

paper. Research has shown that such an observed collaboration marks the beginning of a strong tie that generally lasts beyond the collaboration date. Uniquely determining author identity posed a challenge, as the SCI database uses only the last name and the first initial to identify authors. This proved to be insufficient to disambiguate most authors from China. Thus, we used names, affiliations, addresses, and e-mail addresses to identify authors. Two static snapshots of the coauthorship networks in CAS are shown in Figure 3. As can be seen, the number of authors in the connected components increased significantly from 1995 ($n = 152$) to 2004 ($n = 3,114$).

Network Measurement

The following social network measures were used to determine the status of authors in the network:

- *Betweenness centrality* (BC): A node has a high betweenness centrality if a large number of shortest paths between other nodes pass through it.
- *Degree centrality* (DC): This is based on the intuition that actors who have more direct ties than do other actors may be in advantaged positions.
- *Structural holes*: Structural holes measures identify boundary-spanning individuals. There are two different approaches to calculating structural holes, constrained based (CONS) and redundancy- or efficiency-based (EFF).
- *CONS* is a summary measure that determines the extent to which a node's connections are to others who are connected to each other. EFF for a node measures the proportion of its links that are nonredundant. We implement both these approaches.
- *Random walk betweenness* (RW): This measure is similar to Freeman's betweenness centrality; however, it includes contributions from all paths between nodes (not just the shortest paths). We included this measure because intuitively, this measure may fit the real knowledge diffusion process in coauthorship networks.

Innovation Diffusion Analysis

We used a Cox proportional hazards model for repeated events to study the effect of network measures on paper citation (Nerkar & Paruchuri, 2005; Podolny & Stuart, 1995) and knowledge diffusion. The proportional hazards model was found to be the best fit for our data (We tested the Weibull and exponential models.) The hazard rate model is used because it incorporates information on both censored and uncensored cases (i.e., whether a paper is cited). If T is the duration since a paper has been published, then the hazard rate of a paper being cited again at time t is defined as:

$$r(t) = \lim_{\Delta t \rightarrow 0} \frac{\Pr(t \leq T \leq t + \Delta t)}{\Delta t}$$

Using a semiparametric Cox model, the previous equation can be estimated using the following (Allison, 1995; Cox, 1972):

$$r(t) = h(t) \exp\{XB + Y(t)S\}$$

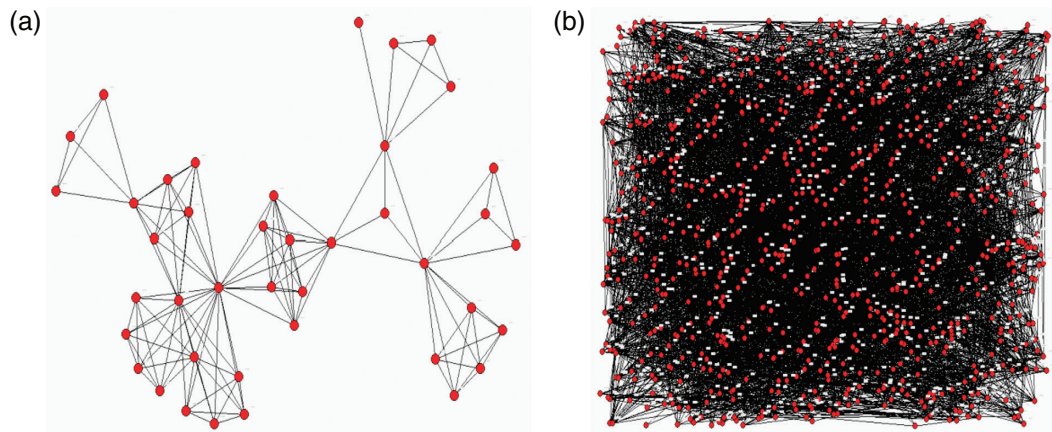


FIG. 3. Coauthorship network (connected component) for the Chinese Academy of Sciences in 1995 (a) and in 2004 (b).

TABLE 3. Variables included in the Cox regression model.

Type	Name
Dependent variable	Paper citation
Control variables	Calendar age
	Paper age
	Keyword count
	Age of prior art
	Self-citation
	No. of references
	Team size
Explanatory variables	Presence of different groups
	Journal effects
	Betweenness centrality
	Degree centrality
	Structural holes
	Random walk

where $r(t)$ is the hazard rate of a paper being cited within the network (dependent variable), $h(t)$ is an unspecified baseline rate, X is a matrix of time-constant covariates (control variables), $Y(t)$ is a matrix of time-varying covariates (independent variables), and B and S are vectors of unknown regression parameters.

The three kinds of variables in our study are discussed in the following sub-sections and summarized in Table 3.

Dependent variable. *Paper citation* was used as the dependent variable because it is considered an excellent measure of technological impact as well as knowledge flow (Ahuja, 2000; Nerkar & Paruchuri, 2005; Podolny & Stuart, 1995). To include time into the analysis, the life of a paper (from publication year until the end of the dataset) was divided into time periods called *spells* (Nerkar & Paruchuri, 2005). The strategy of dividing time into spells effectively measures the effect of network measures on the citation of a patent through its entire life. Each time period was used as a data point to determine the effect of explanatory and control variables on the dependent variable in that spell. In line with prior research (Nerkar & Paruchuri, 2005), spells of a maximum of 1 year

were created for each paper. The first spell begins at the publication year and ends either with a citation or no citation. We calculated the network measures and time-varying covariates for each paper in each spell based on the network of authors in the prior calendar years. This is important because if the social network measures are calculated only once (for a single snapshot of the knowledge network), then it will not accurately reflect the author's status over time. The status of an author might change as the size and density of the network change.

Figure 4 is an example of the spells created for a paper published in 1995 and cited three times in 1996, 1998, and 2002. The solid line denotes that the spell ended with a citation, the dashed line denotes that the spell ended with no citation. The network range indicates the time span of the knowledge network that was used for computing measures for that spell (set to 3 years based on previous research; Podolny & Stuart, 1995).

Explanatory variables. Each of the four social network measures from the previous section [BC, DC, structural holes (EFF and CONS), and RW centrality] were used as explanatory variables to examine the effect of status on paper citation.

Control variables. Control variables are the factors (other than network measures) that affect paper citations. Two kinds of control variables were included based on previous studies (Nerkar & Paruchuri, 2005; Podolny & Stuart, 1995): variables effecting citation rates and variables describing paper attributes. The variables effecting citation rates included the *calendar age*, which controls for the improvement in search technology since the start of the dataset. Information retrieval and search technology have improved dramatically since 1976, and papers are easier to find and cite. Another time-variant control included is the *paper age*. This was included since an older paper may be cited more than are newer papers. Variables describing paper attributes are:

- *keyword count*: Keywords indicate the value of a paper and the technological spaces it occupies.

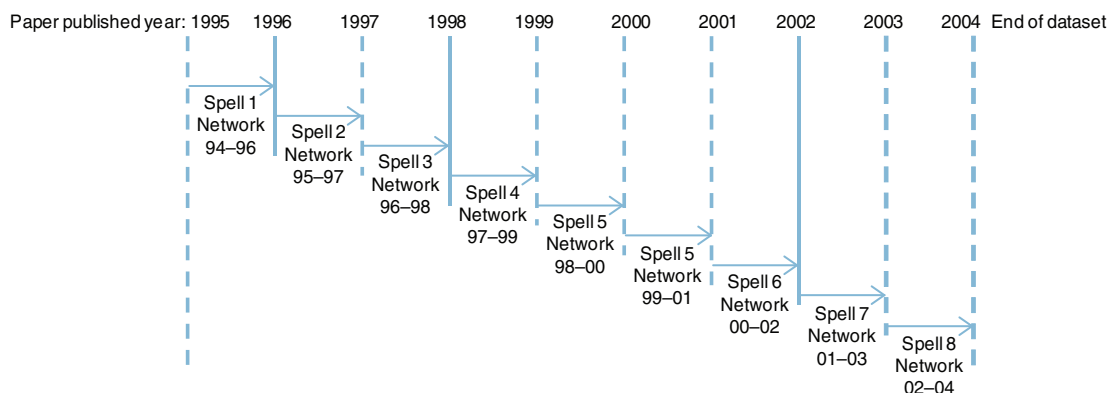


FIG. 4. Spell construction for the dependent variable (paper citation).

- *age of prior art (prior age)*: This indicates that papers that build on old knowledge have different citation patterns than do new ones. This variable is operationalized by the median of the difference between the publication year of a focal paper and that of references.
- *self-citation*: A self-citation indicates confidence of an individual in his or her work and may encourage other individuals to cite that work (Nerkar & Paruchuri, 2005). We set this to be 1 if a paper has been self-cited by any of its authors before the spell.
- *number of references (ref count)*: Papers that cite more prior art may have a different level of influence than do others. We measure this by counting the number of the paper's references.
- *team size*: We include team size to account for the effect of all authors on the paper.
- *presence of different groups (group flag)*: Knowledge may have a higher chance to diffuse if the authors have multiple affiliations. This variable is set to 1 if any author of the paper has a different affiliation from the others.
- *journal effects (journal flag)*: This variable is included to control for the different influential journals. It is implemented as a dummy variable that captures the top-10 journals (selected by impact scores). A value of 1 indicates that the paper is published in top10 journals.

Based on the aforementioned variables, the Cox model tests the effect of an author's status (as established by a network measure) on the likelihood of his or her knowledge being selected for citation by other authors. Once the appropriate measure is determined, we identify the most influential authors in nanotechnology for each of the three institutions.

Results

Here, we show the results of the Cox proportional hazards analysis for the CAS, the RAS, and the IIT. Six models were constructed for these three institutions; one included only control variables, and the other five included all control variables and each social network measure individually. Results for the CAS are shown in detail; the conclusions for the RAS and the IIT are discussed in summary when they are similar to those of the CAS.

Knowledge Diffusion Analysis: The CAS

The correlation matrix with all the variables was examined to help identify multicollinearity and redundancy issues. The results showed that all correlations were low and did not pose any problems (see values in Table 4). The high correlations between some network measures (in bold) do not lead to multicollinearity issues since each model contains only one measure.

Table 5 presents the Cox proportional hazards regression model for the CAS. Numerals in the columns represent the hazard ratio (HR) (Significance at $p < 0.01$ level is noted with an asterisk.) Model 0 consists of all the control variables. As can be seen, the likelihood of a paper being cited decreases ($HR < 1$) with an increase of the number of keywords. The likelihood of citation also decreases when the authors have different affiliations. This was unexpected; however, it might be an artifact of our methodology. Since we only measure citation within institutions, different affiliations may indicate knowledge diffusion across organizations, which is not measured in our model. On the other hand, the likelihood of a paper being cited increases ($HR > 1$) with an increase of reference counts and author counts. Papers that cite more prior work and have more authors may have more influence and are much more likely to be cited. Results also showed that the presence of self-citation and being published in top journals positively affect the citation of the paper. The significance of these control variables generally persisted across all models.

Model 1 adds Freeman's BC with all the control variables. As can be seen from Table 5, the BC measure of the author was found to have a significant positive effect on the citation of his or her papers. However, the HR was almost equal to 1 (1.000004), indicating that the measure cannot suitably explain the effect of an author's status on the selection of his or her knowledge for innovation in the CAS.

Model 2 adds DC to the control variables in Model 0. The DC of an author was found to have a significant positive effect on the citation of his or her papers, but also with a HR slightly larger than 1 (1.005815). This also implies that the measure may not be suited to explain the effect of an author's status on the selection of his or her knowledge in the CAS.

TABLE 4. Correlation matrix between control variables (Chinese Academy of Sciences).

Variables	1	2	3	4	5	6	7	8	9	10	11	12	13	14
1. BC	1													
2. DC	0.87*	1												
3. EFF	0.59*	0.73*	1											
4. CONS	0.23*	0.36*	0.79*	1										
5. RW	0.09*	0.16*	0.46*	0.62*	1									
6. Prior age	0.02*	0.01	−0.03*	−0.06*	−0.06*	1								
7. Keyword count	−0.02*	−0.04*	−0.00	0.00	−0.00	0.00	1							
8. Self-citation	0.06*	0.06*	−0.02*	−0.02	−0.04*	0.45*	−0.04*	1						
9. Reference count	−0.07*	−0.08*	−0.11*	−0.10*	−0.06*	0.11*	−0.03*	0.16*	1					
10. Team size/ Author count	0.10*	0.12*	0.03*	0.03*	0.04*	0.03*	−0.04*	0.02*	−0.05*	1				
11. Group flag	−0.06*	−0.05*	−0.02*	0.05*	0.04*	−0.01	−0.03*	0.10*	−0.01	0.21*	1			
12. Journal flag	−0.02*	−0.03*	−0.05*	−0.05*	−0.04*	0.01	−0.02*	0.03*	0.12*	0.02*	−0.21*	1		
13. Paper age	0.24*	0.18*	0.17*	0.08*	−0.03*	−0.15*	−0.07*	−0.07*	−0.07*	−0.02*	0.05*	0.06*	1	
14. Calendar age	0.14*	0.01	−0.13*	−0.20*	−0.23*	0.15*	0.07*	0.07*	0.07*	0.02*	0.05*	−0.06*	0.20*	1

BC = betweenness centrality; DC = degree centrality; EFF = redundancy or efficiency-based; CONS = constrained-based; RW = random walk.

* $p < 0.05$.

TABLE 5. Knowledge diffusion results for the Chinese Academy of Sciences.

Variables	Model 0	Model 1	Model 2	Model 3a	Model 3b	Model 4
BC		1.000004*				
DC			1.005815*			
EFF				1.359633*		
CONS					1.326059*	
RW						1.25256
Prior age	0.9980615	0.9979607	0.9979044	0.9970947	0.9991697	0.9985351
Keyword count	0.9078333*	0.9084321*	0.9084989*	0.9060713*	0.9063287*	0.9077845*
Self-citation	1 class* (1.360126)	1 class* (1.339293)	1 class* (1.333197)	1 class* (1.334097)	1 class* (1.354752)	1 class* (1.362013)
Reference count	1.006981*	1.007228*	1.00726*	1.007497*	1.007288*	1.007052*
Author count	1.05373*	1.048461*	1.046922*	1.051576*	1.053564*	1.052994*
Group flag	1 class* (0.7105347)	1 class (0.7234883)	1 class	1 class* (0.7164366)	1 class (0.7027412)	1 class* (0.7073208)
Journal flag	1 class* (2.387818)	1 class* (2.399323)	1 class* (2.450417)	1 class* (2.536319)	1 class* (2.504112)	1 class* (2.434536)
Paper age	1.882723	1.635927	1.089369*	1.205402	1.702875	2.988541
Calendar age	1.058284	1.05783*	1.065479	1.080535*	1.075622	1.064521*

Note. The self-citation, group, and journal flags were split into dummy classes for the analysis. BC = betweenness centrality; DC = degree centrality; EFF = redundancy or efficiency-based; CONS = constrained-based; RW = random walk.

* $p < 0.01$.

TABLE 6. Top-5 authors in the Chinese Academy of Sciences.

Rank	EFF	CONS
1	B. Zou, Inst Phys, Beijing (1999–2001)	D. Jiang Shanghai, Inst Ceram, State Key Lab High Performance Ceram & Superfine, Shanghai (2002–2004)
2	J. Yuan, Inst Chem, Ctr Mol Sci, Lab Chem Biol, Beijing (2001–2003)	T. Zhao, Inst Chem, Ctr Mol Sci, State Key Lab Engn Plast, Beijing (2002–2004)
3	D. Shi, Beijing Lab Vacuum Phys, Beijing (2002–2004)	Q. Zhang, Inst Phys, Opt Phys Lab, Beijing (2002–2004)
4	H. Cong, Inst Met Res, Shenyang Natl Lab Mat Sci, Shenyang (2000–2002)	W. Hu, Inst Chem, Ctr Mol Sci, Beijing (2002–2004)
5	G. Fei, Inst Solid State Phys, Key Lab Mat Phys, Hefei (2001–2003)	Q. Zhang, Inst Phys, Opt Phys Lab, Beijing (2001–2003)

EFF = redundancy or efficiency-based; CONS = constrained-based.

Model 3a and 3b consists of the two structural holes metrics along with all the control variables. EFF was found to be positively significant, with an HR of 1.359633. This implies that a unit increase in the EFF measure of an author leads to a 35% increase in the likelihood of a citation of his or her paper. CONS also was found to be significant. A unit increase in the CONS measure of an author leads to a 32% increase in likelihood of a citation of his or her paper.

Model 4 tests the HR for RW and all control variables. It was found that the RW measure was positively related, but not significant at a 1% level.

In summary, structural holes (EFF and CONS) were suited to explain the status of authors in the knowledge diffusion process in the CAS as compared to the other measures. This supports previous research on inventor status in patent networks (Nerkar & Paruchuri, 2005).

Top Authors in the CAS

Since the EFF and CONS status measures more clearly explain the knowledge diffusion pattern in the CAS, we report on the most prestigious authors as identified by these measures in Table 6. For instance, Bingsuo Zou, Inst Phys, Beijing (1999–2001) in the EFF column implies that based on the EFF measure, Bingsuo Zou was the most important author in the 2-year time span of 1999 to 2001. We found that the most prestigious authors in different subject categories such as physics, chemistry, and material science were from Beijing.

TABLE 7. Knowledge diffusion results for the Russian Academy of Sciences.

	BC	DC	EFF	CONS	RW
Direction	Positive	Positive	Positive	Positive	Positive
Hazard Ratio	1.000157	1.014428*	2.303907*	2.069452*	2.179456*

BC = betweenness centrality; DC = degree centrality; EFF = redundancy or efficiency-based; CONS = constrained-based; RW = random walk.

* $p < 0.01$.

TABLE 8. Top-5 authors in the Russian Academy of Sciences.

Rank	EFF	CONS	RW
1	V.Y. Prinz, Inst Semicond Phys, Novosibirsk (2002–2004)	R.A. Andrievskii, Inst Problems Chem Phys, Chernogolovka, Moscow Oblast (2000–2002)	V.Y. Prinz, Inst Semicond Phys, Novosibirsk (2002–2004)
2	B.V. Volovik & A.F. Ioffe, Phys Tech Inst, St. Petersburg (2002–2004)	W. Skorupa, Inst Semicond Phys, Siberian Div, Novosibirsk (2002–2004)	L.M. Bronstein & A.N. Nesmeyanov, Organoelement Cpd Inst, Moscow (2002–2004)
3	V.D. Frolov, Inst Gen Phys, Moscow (2002–2004)	R.A. Andrievskii, Inst Problems Chem Phys, Chernogolovka, Moscow Oblast (1999–2001)	P.M. Valetsky & A.N. Nesmeyanov, Organoelement Cpd Inst, Moscow (2002–2004)
4	S.M. Pimenov, Inst Gen Phys, Moscow (2000–2002)	E.D. Obraztsova, Inst Gen Phys, Moscow (2000–2002)	V.Y. Prinz, Inst Semicond Phys, Novosibirsk (2001–2003)
5	L.M. Bronstein & A.N. Nesmeyanov, Organoelement Cpd Inst, Moscow (2002–2004)	G.V. Kalinnikov, Inst Problems Chem Phys, Chernogolovka, Moscow Oblast (2000–2002)	V.V. Preobrazhenskii, Inst Semicond Phys, Siberian Div, Novosibirsk (2002–2004)

Knowledge Diffusion Analysis: The RAS

The results for the RAS are summarized in Table 7. All measures except BC were found to have a significant positive effect on the citation of papers (although the HR for DC was 1.014428). The HRs showed that the structural holes measures and RW explained the knowledge diffusion pattern within the RAS. One unit increase in EFF leads to a 130% increase in the hazard of citation as compared to 118% for RW and 107% for CONS.

Top Authors in the RAS

Table 8 lists the most important authors in the RAS using the three effective and statistically significant measures. The measures demonstrate the importance of the same authors in different spells. For instance, Prinz, VY was identified by EFF and RW to have high importance in all time spells from 2001 from 2004. Andrievskii, RA also was high status as demonstrated by CONS in the spells from 1999 to 2002. Bronstein, LM also was ranked in both EFF and RW's top author lists. Most of these authors were from physics and related areas, and the majority were from the cities of Moscow and Novosibirsk.

Knowledge Diffusion Analysis: The IIT

Table 9 shows a summary of the results obtained for the IIT in which all the measures were found to have a significant positive effect. However, the RW measure best explained the knowledge diffusion (with 215% increase in the hazard of citation), followed by EFF and CONS.

Top Authors in the IIT

Using the most effective measures, RW, EFF, and CONS, we identified the most important authors in the IIT (Table 10). RW identified Gajbhiye, NS's importance in the knowledge diffusion process from 1997 to 2001; EFF identified Das, RN's important role in all spells from 2000 to 2004. The majority of those authors were from chemistry- and

physics-related branches; however, there was no overlap in the authors identified by the measures.

Comparing results across the three institutions, it is clear that boundary-spanning individuals (as measured by structural holes) are considered influential in the CAS and the RAS whereas in the IIT, most influential individuals were identified by RW betweenness. In addition, it appears that the important authors in the IIT persisted across longer spells in time as compared to those in the RAS and the CAS. Another important conclusion that can be drawn from the results is that being connected to diverse parts of the network leads to more influence since both structural holes and RW are essentially betweenness measures emphasizing “gatekeeper” roles.

We plan to closely examine the data and networks extracted, as the significance of different measures indicate that cultural and institutional differences may influence knowledge diffusion. In addition, each measure makes implicit assumptions on the flow of knowledge in the network. The significance (or lack thereof) of these measures may indicate differences in institutional and collaborative culture. Further study on the networks from India may be needed because the current dataset is significantly smaller than the datasets of China and Russia.

Conclusions and Future Directions

In this study, we examined the role of author status on the selection of knowledge that is recombined to produce

innovation within an organization. We conducted the study on three large-scale institutions from three different countries and studied the process that models their knowledge evolution in nanotechnology fields. Using different social network measures, we identified the mechanism through which knowledge is selected for recombination. Specifically, we asked how the status of an individual in a coauthorship network affects knowledge diffusion. To answer this, we used Cox proportional hazards models to examine the role of author status on the selection of knowledge that is recombined to produce innovation within an organization. The empirical results showed that individuals with access to diverse parts of the knowledge network were influential in all three institutions.

Our second research question focused on which measures could explain the effect of an individual’s status on knowledge diffusion. Since Cox proportional hazards models can be used to compare the effect of a set of variables on a future event, by measuring individual status in coauthorship networks with different social network measures (BC, DC, structural holes, RW betweenness), we can identify the most effective measure to determine author status for a specific institution. For example, our empirical study showed that for the CAS in China, structural holes (as measured by the efficiency measure) best explained the status of authors in the knowledge diffusion process in China. Using the most suited measures in each institution, we also identified the most important authors in the nanotechnology field in China, Russia, and India.

There are some limitations to this study. Although we found that cultural and institutional differences may result from the varying knowledge diffusion mechanisms within institutions, a further sociological study to understand which cultural factors might influence diffusion is warranted. Thus, comparing institutional culture and incentives in these countries and its effect on network structure (and the selection of knowledge) might be an interesting future direction to pursue. We also plan to compare these results to organizations in the United States and European nations to examine any differences. In addition, since paper collaborations are only a subset of actual affiliations between individuals in

TABLE 9. Knowledge diffusion results for the Indian Institutes of Technology.

	BC	DC	EFF	CONS	RW
Direction	Positive	Positive	Positive	Positive	Positive
Hazard ratio	1.01309*	1.09702*	2.929148*	2.113073*	3.150772*

BC = betweenness centrality; DC = degree centrality; EFF = redundancy or efficiency-based; CONS = constrained-based; RW = random walk.

* $p < 0.01$.

TABLE 10. Top-5 authors in the Indian Institutes of Technology.

Rank	EFF	CONS	RW
1	N. Sandhyarani, Dept Chem and Reg Sophisticated Instrumentat Ctr, Madras (2001–2003)	R.K. Joshi, Dept Phys, New Delhi (2002–2004)	N.S. Gajbhiye, Dept Chem, Kanpur, Uttar Pradesh (1998–2000)
2	N. Pradhan, Dept Chem, Kharagpur (2002–2004)	H.K. Sehgal, Dept Phys, New Delhi (2002–2004)	N.S. Gajbhiye, Dept Chem, Kanpur, Uttar Pradesh (1997–1999)
3	R.N. Das, Dept Chem, Kharagpur 721302, W Bengal (2002–2004)	Y.V.G.S. Murti, Dept Phys, Guwahati (2000–2002)	N.S. Gajbhiye, Dept Chem, Kanpur, Uttar Pradesh (1999–2001)
4	R.N. Das, Dept Chem, Kharagpur 721302, W Bengal (2000–2002)	P. Nandakumar, Dept Phys, Chennai (2000–2002)	B.R. Mehta, Dept Phys, Thin Film Lab, New Delhi (2000–2002)
5	M. Venkataramanan, Dept Chem and Reg Sophisticated Instrumentat Ctr, Madras, Tamil Nadu (2000–2002)	C. Vijayan, Dept Phys, Chennai (2000–2002)	K.C. Khilar, Dept Chem Engn, Bombay, Maharashtra (2002–2004)

EFF = redundancy or efficiency-based; CONS constrained-based; RW = random walk.

an institution, patent collaborations and other kinds of relationships can be overlaid over paper collaborations to more clearly understand individual status.

Our findings also suggest that authors are in a position to work toward prestige positions in knowledge networks to play important roles in an organization's knowledge evolution. This may be even more important in industry research labs, where the evolution of knowledge and the patents that it is based on may have important financial implications. Policy makers in the scientific community should pay more attention to the network positions and knowledge diffusion mechanisms because the development of an organization's capabilities and competencies might depend on the path of knowledge flow and the individuals on that path. The success of betweenness measures of prestige shows that the nanotechnology community and possibly the scientific community at-large could benefit by encouraging collaborations between diverse parts of the discipline.

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